B Open Domain vs In-Domain

As explained in prior work (Makelov et al., 2024), it remains unclear whether the dataset used for training the SAE should come from the same domain as the evaluation data (In Domain) or from a general dataset (Open Domain). In this section, we evaluate the SAEs trained on the WiC dataset (Pilehvar & Camacho-Collados, 2019) as the In Domain dataset and Red Pajama (Weber et al., 2024) as the Open Domain dataset. Figure 8 shows the results, where recall is higher for the Open Domain across all expand ratios, while precision is higher for models trained on the In Domain dataset.

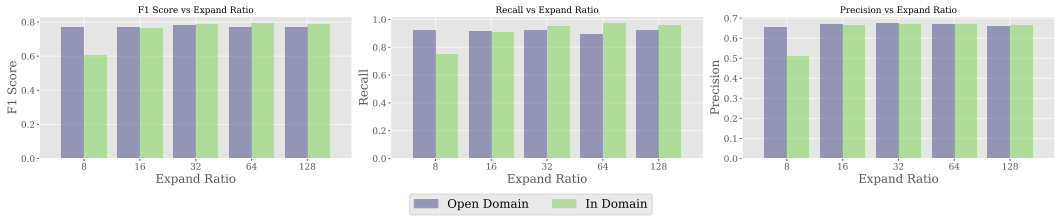

Figure 8: Comparison of F1 Score, Recall, and Precision across different Expand Ratios (32, 64, 128) for Open Domain (red pajama (Weber et al., 2024)) and In Domain (WiC data) settings. The results show that Open Domain outperforms In Domain in terms of recall, while In Domain performs better in terms of precision.

C Details of Experiment

Here are the detailed parameters for this experiment. We initialize the weights of the Decoder as the transposed weights of the Encoder, following the methodology from previous studies (Gao et al., 2024).

Table 4: Summary of Training Configuration and Hyper-parameters.

Parameter	Value
Batch Size	8192
Total Training Steps	200,000
Learning Rate	2e-4
Input Dimension (d_{in})	768 (GPT2-small)
Context Size	256

D Random SAE

In this section, we analyze the performance of PS-Eval when SAEs are activated randomly. The experimental setup assumes an expand ratio of 32, meaning that the number of features in the SAE is calculated as expand ratio $\times\ d_{\text{model}=32\times768=23576}$. We compute the probability of overlapping activations (P_{same}) and the probability of non-overlapping activations (P_{diff}).

The total number of ways to choose 2 features from 24576 features is given by:

$$\binom{24576}{2} = \frac{24576 \times (24576 - 1)}{2}$$

The probability of selecting the **same** feature is:

$$P_{\text{same}} = \frac{1}{\frac{24576\times(24576-1)}{2}} = \frac{2}{24576 \times 24575} = 3.3115039e - 9$$

The probability of selecting the **different** feature is:

$$P_{\text{diff}} = 1 - P_{\text{same}} = 0.99999999668$$

Next, we calculate the True Positives (TP), False Negatives (FN), False Positives (FP), and True Negatives (TN). We assume that the total number of data samples is $N = 1112$, and the number of samples with the same meaning is $n = 556$.

- True Positives (TP):
$$\text{TP} = n \times P_{\text{same}} \approx 0.00000184119$$
- False Negatives (FN):
$$\text{FN} = n \times P_{\text{diff}} \approx 555.999998154$$
- False Positives (FP):
$$\text{FP} = (N - n) \times P_{\text{same}} \approx 0.00000184119$$
- True Negatives (TN):
$$\text{TN} = (N - n) \times P_{\text{diff}} \approx 555.999998154$$

The metrics for the random activation of SAEs are summarized in Table 5.

Table 5: Summary of evaluation metrics with their respective formulas and calculated values. These metrics highlight the model's performance in terms of accuracy, precision, recall, specificity, and F1 score.

Metric	**Formula and Value**
Accuracy	$\text{Accuracy} = \frac{\text{TP}+\text{TN}}{N} = 0.5$
Precision	$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}} = 0.5$
Recall (Sensitivity)	$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} \approx 0.0000000319$
Specificity	$\text{Specificity} = \frac{\text{TN}}{\text{TN}+\text{FP}} \approx 0.99999999668$
F1 Score	$\text{F1} = \frac{2\times\text{Precision}\times\text{Recall}}{\text{Precision}+\text{Recall}} \approx 0.0000000638$

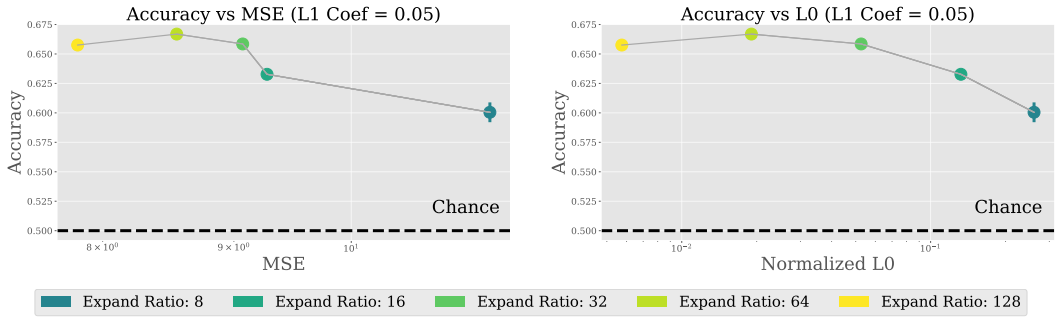

Figure 9: Comparison of Accuracy vs MSE and L_0 for different Expand Ratios (L_1 Coef = 0.05). The left panel shows the relationship between accuracy and mean squared error (MSE), while the right panel presents accuracy versus normalized L_0 sparsity. The markers represent various expand ratios (8, 16, 32, 64, and 128). Higher expand ratios generally show a trend towards better performance, as indicated by the yellow and green markers corresponding to expand ratios of 64 and 128. The horizontal dashed line represents the chance rate, providing a baseline for comparison.

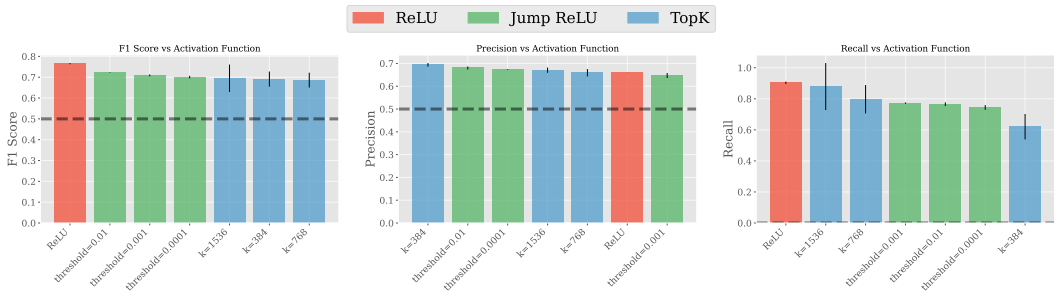

Figure 10: Comparison of F1 score, Precision, and Recall across different activation functions. The models were trained with JumpReLU (with varying thresholds), TopK (with different k), and standard ReLU. The results imply that ReLU may be still better than JumpReLU or TopK with the polysemous words. The error bars indicate the variation across different 3 seeds. Results for smaller k of TopK and the outcomes of JumpReLU using STE are provided in Appendix D. The horizontal dashed line represents the chance rate, providing a baseline for comparison.

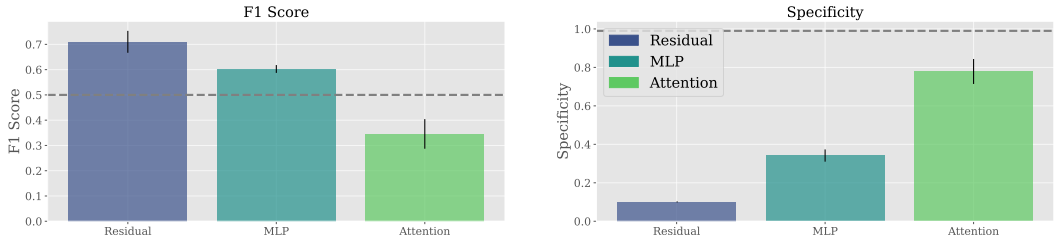

Figure 11: Comparison of F1 score and Specificity for SAEs trained on the outputs of the Residual, MLP, and Attention components. The left panel shows the F1 score, where Residual and MLP achieve higher values, while the Attention output performs lower. In contrast, the right panel shows Specificity, where the Attention output exhibits a significantly higher Specificity, despite the lower F1 score (see Appendix I for details on these components). The horizontal dashed line represents the chance rate, providing a baseline for comparison.

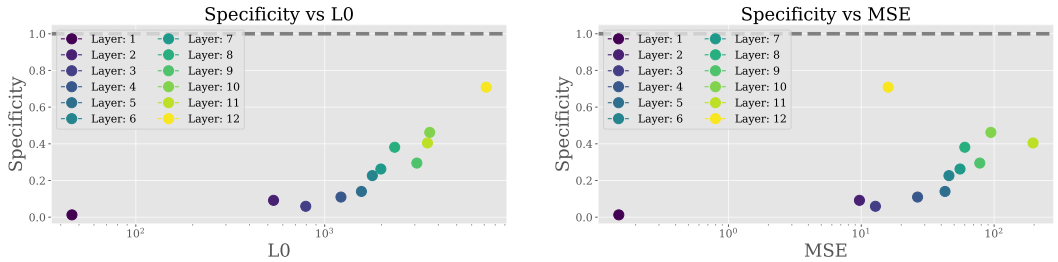

Figure 12: Comparison of Specificity across different layers of the SAE in relation to L_0 (left) and MSE (right). Each point represents a layer, from layer 1 to layer 12, with deeper layers (i.e., layer 6 onwards) showing an increasing trend in Specificity. This indicates that deeper layers exhibit more distinct activations for polysemous words, consistent with previous findings that deeper layers in SAEs face greater challenges in training, leading to higher MSE and L_0 scores. The horizontal dashed line represents the chance rate, providing a baseline for comparison